TINKU GARG

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SUMMARY

Self-driven Machine Learning Engineer with a bias for building over theorizing. Experienced across the full ML stack — time-series forecasting for space weather, physics-based heat-risk modeling, and multi-stage classification for agricultural decision support. Comfortable owning projects end-to-end: from data wrangling and feature engineering through model architecture, class-imbalance strategies, GPU infrastructure, and open-source release. Seeking roles where scientific rigor meets production-grade engineering.

EDUCATION

- Master of Computer Application (MCA) in AIML — **Bennett University** | Expected 2027 | CGPA: 7.00
- Bachelor of Computer Application (BCA) in AIML (IBM) — **JECRC University** | 2025 | CGPA: 7.47
- Higher Secondary (12th) — **U.S.S Senior Secondary School** | 2020 | 74.4%
- Secondary (10th) — **A.V.M Secondary School** | 2018 | 75.17%

WORK EXPERIENCE

Applied AI Research Intern — National Institute of Technology, Delhi | Certificate *Jun 2026 – Jul 2026*

- Designed and trained **Dynoflare-net**, a Temporal Convolutional Network with multi-head attention for binary solar flare prediction (M/X-class vs. quiet), achieving **TSS 0.8089** and **AUC 0.9652** on ~336K multivariate time-series samples from the SWAN-SF benchmark.
- Engineered **108 features** from **54 magnetic-field features**, applying domain-informed feature selection to improve model generalization on large-scale space weather datasets.
- Addressed severe class imbalance (flare events **<5%**) through per-epoch threshold tuning and cosine annealing LR scheduling, improving minority-class recall by **~13 percentage points**.
- Migrated training pipeline from Google Colab to a headless **NVIDIA RTX 4000 Ada** GPU server, resolving multiprocessing and dependency issues for reproducible remote execution.
- Published complete codebase as an **open-source repository on GitHub (Apache 2.0)**. Manuscript in preparation for peer-reviewed journal submission.

ML Engineer Intern — Codestring | Certificate *Feb 2025 – Jun 2025*

- Built an **animal image captioning** system using a **Vision Transformer (ViT) encoder** and **GPT-2 decoder** with Hugging Face Transformers and PyTorch.
- Fine-tuned the vision-language model on animal images paired with captions, building image preprocessing, augmentation, and caption tokenization pipelines.
- Tuned decoding strategies (beam search, length penalty) and packaged inference into a reusable module that generates captions for new images.

PROJECTS

Solar Flare Prediction — Dynoflare-net — GitHub Repo

- Architected an end-to-end **PyTorch** pipeline for M-class and above solar flare prediction using SHARP magnetic-field parameters from the SWAN-SF benchmark.
- Evaluated model performance using **TSS, HSS, AUC, Precision, and Recall**, the standard metric suite for solar physics forecasting research.
- Structured the codebase into modular, reusable components to support extension toward multi-class flare severity prediction.
- Documented the full methodology and released trained model weights publicly for community benchmarking and reproducibility.

Heatwave Early Warning & Human Thermal Stress Index — GitHub Repo

- Engineered a physics-based **Human Thermal Stress Index** (WBGT + UTCI + Heat Index) for Smart India Hackathon 2026, replacing single-threshold dry-bulb alerts with a multi-metric heat-risk model.
- Built a satellite downscaling pipeline using **Google Earth Engine** (MODIS/Landsat LST) to generate ward-level heat-island offsets for Ahmedabad.
- Validated against 4 historical pre-monsoon seasons (492 days, ERA5 data): zero missed severe episodes and **3–9 days advance warning**.
- Built a **FastAPI** GIS dashboard with ward-level alerts and an offline fallback mode, backed by **87 automated tests**.

Fruit Harvesting Intelligence System — GitHub Repo

- Constructed a two-stage deep learning pipeline combining **CNN feature extraction** with an **ANN harvest-decision model** using TensorFlow and OpenCV.
- Achieved **19-fruit species identification** through robust image preprocessing and augmentation techniques.
- Implemented 3-class ripeness classification (Ripe/Unripe/Damaged) to support automated quality assessment.
- Designed the system to output harvest-timing recommendations, supporting real-world agricultural decision-making.

TECHNICAL SKILLS

Languages: Python, SQL
ML/DL Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NumPy, Pandas, OpenCV
Techniques: CNN, LSTM, TCN, Attention Mechanisms, Transfer Learning, Data Augmentation, Class-Imbalance Handling, Feature Engineering
Tools & Infra: Git, GitHub, Linux (headless GPU server), Docker, NVIDIA RTX 4000 Ada, Google Colab, Matplotlib, FastAPI, Earth Engine

CERTIFICATIONS

- **IBM:** Deep Learning Fundamentals | Machine Learning with Python | Predictive Modeling (SPSS) | Cognitive Decision System for Managers | Watson Studio | Rapid Development for AI | Text Analytics 101 | Data Visualization | Introduction to Python | Introduction to AI
- **Google:** Foundations — Data, Data, Everywhere | Connect and Protect — Networks and Network Security
- **Coursera:** SQL for Data Science (UC Davis) | Text Retrieval and Search Engines (UIUC)
- **Intel:** Introduction to Generative AI
- **Infosys:** Step-by-Step ML with Python | Introduction to Hardware & Software | 3D Modeling
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